Corto Física

1) $d_1 = 10cm$ $\quad 10cm * \frac{1m}{100cm} = 0.1m$

$P_1 = 8 \times 10^4 Pa$

$d_2 = 5cm$ $\quad 5cm * \frac{1m}{100cm} = 0.05m$

$P_2 = 6 \times 10^4 Pa$

Caudal

$Q = AV$ $\quad A_1 = \frac{\pi d_1^2}{4} = \frac{\pi (0.1)^2}{4} = 7.85 \times 10^{-3} m$

$A_2 = \frac{\pi d_2^2}{4} = \frac{\pi (0.5)^2}{4} = 1.96 \times 10^{-3} m$

Velocidad

$P = \frac{\rho V^2}{2}$ $\quad V_1 = \sqrt{\frac{2P}{\rho}}$

$V = \sqrt{\frac{2(8 \times 10^4)}{997 \, Kg/m^3}} = 160.48 m/seg$

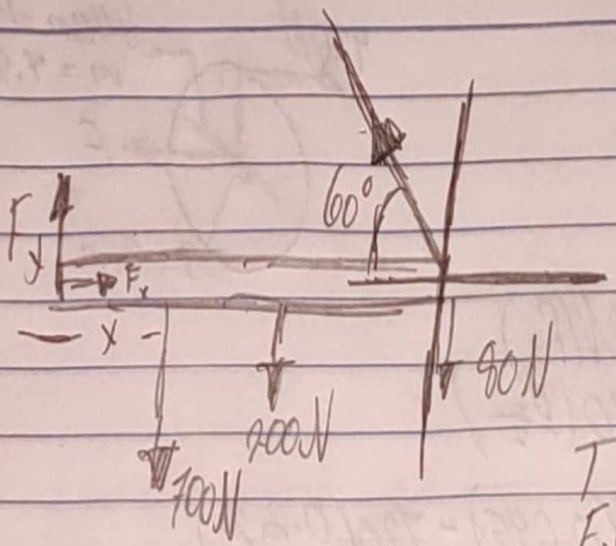
2) $d = 80cm$ $\qquad w = 900\ rev/min$ $\qquad \theta = 37^\circ$

$w = \frac{\Delta\theta}{t}$ $\qquad V = \frac{d}{t}$

$t = \frac{\Delta\theta}{w}$ $\qquad V = \frac{wd}{\Delta\theta}$

$$V = \frac{900 \frac{\cancel{rev}}{\cancel{min}} * \frac{360^\circ}{1\cancel{rev}} * \frac{1\cancel{min}}{60s} * 0.8m}{37^\circ} = 1394 m/s$$

4) a)



$-F_x - T\cos 60° = 0$

$F_y - 980N + T\,sen60° = 0$

$\Sigma \tau = -700N(1m) - 200N(3m) - 80N(6m) + T\,6\,sen60° = 0$

$T = 342.6N$

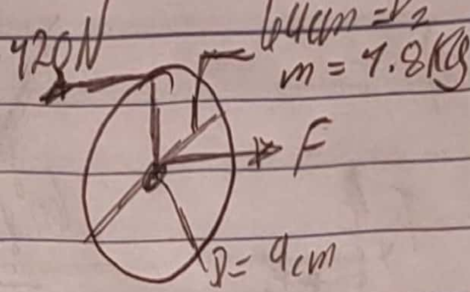
$F_y = 683.3N$

$F_x = 171.3N$

b) $\Sigma \tau = -700N(x) - 200N(3m) - 80N(6m) + 900N(6)\,sen60° = 0$

despejando X $\quad x = 5.13m$

5)



$$\tau_R = \tau_F - \tau_{Fr}$$

$$I\alpha = F \cdot R_1 - 720N(R_2)$$

$$(mR^2)\alpha = F\left(\frac{D_1}{2}\right) - 720\left(\frac{D_2}{2}\right)$$

$$(7.8)(0.32m)^2(4.5) = F(0.045) - 720(0.32)$$

$$0.829\,\frac{kg\,m^2}{s^2} = F(0.045) - 38.4\,Nm$$

$$0.829 + 38.4 = F(0.045)$$

$$\frac{39.229}{0.045} = F$$

$$\underline{F = 871.76N}$$